Measured cost of cosine-filter ground-state preparation from a matrix-product state: gap, overlap and target in the J_1–J_2 chain, with a comparison to QETU

Authors to be added
(Dated: November 3, 2026)

A common route to a ground state on a quantum computer compiles a classical trial state and removes its residual excited weight with a quantum filter. We measure what this costs by exact simulation of the Trotterized circuits for the spin-$\frac{1}{2}$ J_1–J_2 model, with a matrix-product-state (MPS) trial and the one-ancilla cosine filter of Stetcu, Baroni and Carlson (SBC). The central result is a controlled separation of the two inputs that the known complexity of filtering names. Holding the gap Δ fixed by tuning J_2 and the trial overlap γ fixed by modifying the trial, at $N = 8$–14, the filter's CNOTs per shot grow as $\Delta^{-1.91}$ (95% interval 1.60–2.16) but only as $\gamma^{-0.14}$, while its cost per success grows as $\gamma^{-1.14}$: the gap sets the circuit and the overlap sets the repetitions. Two different ways of lowering the overlap, a sublattice rotation and an entanglement truncation, give the same law. The long, smooth SBC pulses tolerate Trotter steps of order $1/J_1$, a median 51 times coarser than the commutator bound for the same circuit. Against a Chebyshev (QETU) filter under identical accounting, SBC is cheaper at $\epsilon \approx 10^{-2}$, the two tie at 10^{-3}, and QETU is cheaper from 10^{-4}; because the QETU step shrinks as $1/N$, the crossover target tightens with size, from $\epsilon \approx 6.0 \times 10^{-3}$ at $N = 8$ to 6.4×10^{-4} at $N = 14$. Filtering a cheap MPS beats compiling a better one on chains only as a last-mile polisher (18 of 63 targets), but at 27 of 28 targets on 4×3 and 4×4 cylinders, and still at 25 when every MPS circuit is charged half its cost. Planned from DMRG energies, with the Trotter step certified on the device by the Eckart bound, every certified state met its target in the 91 cases checked against exact diagonalization; overestimating the gap or the overlap is what breaks a filter, and a 10% overlap margin removes the overlap failures.

I. INTRODUCTION

A common route to a ground state on a quantum computer starts classically: an approximate method produces a trial state that can be compiled into a short circuit, and a quantum subroutine removes its residual excited-state weight. For one-dimensional systems the natural classical engine is DMRG [2, 3], which yields a matrix-product state (MPS) whose low-bond-dimension truncations compile exactly into staircase circuits [4]. The quantum step can be phase estimation [5], a polynomial filter implemented by quantum singular-value transformation [6–8], or a product of cosine filters implemented with a single ancilla [1, 12]. The last option, the projection algorithm of Stetcu, Baroni and Carlson (SBC) [1], needs little beyond time evolution: each pulse is one controlled time evolution between ancilla rotations and a measurement.

Resource estimates for such filters usually rest on worst-case bounds, which are loose by orders of magnitude for Trotter errors [29]. Here we measure instead. For the open J_1–J_2 Heisenberg chain and small cylinders we simulate every circuit exactly and find, for each case, the shortest filter and the coarsest Trotter step that reach a target infidelity ϵ. Our main question is *what sets the cost.* The spectral gap Δ and the trial overlap γ normally change together with system size N, so the textbook statement that filter depth scales as $1/\Delta$ and repetitions as $1/\gamma$ [5, 6] has not been tested on real circuits with the two varied separately. We decouple them, holding the gap fixed by tuning J_2 and the overlap fixed by modifying the trial in two different ways, and fit the measured costs with error bars. We then ask *which filter is cheaper*, comparing SBC with a Chebyshev polynomial filter (QETU) [8] under identical accounting, and *when filtering pays*, comparing filtering a cheap MPS with compiling a better one. Finally, we run the workflow from DMRG inputs with a Trotter step certified on the device, and test its sensitivity to wrong inputs, on systems where exact diagonalization verifies the result.

Relation to prior work. The filter is SBC's [1]; it shares its circuit with the rodeo algorithm [12, 20], which draws the times at random. Our "dyadic" schedule, a halving ladder of times, is the one SBC and Rule *et al.* use for quantum-number projection [1, 21], and for energy it is a finite rodeo super-iteration [22, 23]. That filter depth scales as $1/\Delta$ and repetitions as $1/\gamma$ is the known complexity of filter methods [5, 6]. QETU has been costed with Trotterized evolution [24] and run on small Hubbard models [25, 26]. Resource studies that compile a classical trial and refine it exist for quantum chemistry [10, 11, 27], and Watanabe *et al.* [28] finish an early-stopped MPS encoder with probabilistic imaginary-time evolution on Heisenberg chains. What is new here is (i) a controlled separation of gap and overlap at fixed size, with two overlap mechanisms, which turns the known scaling into measured exponents with confidence intervals; (ii) a measured factorization of the cost into filter length, Trotter step and gates per step, with the measured step set against worst-case and low-energy Trotter bounds; (iii) a like-for-like measured comparison of cosine and Chebyshev filters, including the Chebyshev filter's query length; (iv) a measured benchmark of filtering a cheap MPS against compiling a better one, with harder baselines; and (v) DMRG-planned runs with a certified step rule and sensitivity studies, verified against exact diagonalization. The code and data are at `https:`

//github.com/aaronseymour7/FilterPipeline.

II. FILTERS, CIRCUITS AND HOW WE MEASURE THEM

Spectral window. With exact energies E_0, E_1 and an upper bound $E_{\max}$ on the spectrum, the scaled Hamiltonian $H_s = (H - E_0)/W$, $W = E_{\max} - E_0$, has its ground energy at 0 and every other level in $[\delta, 1]$, $\delta = \Delta/W$. Throughout, $\Delta = E_1 - E_0$ is the gap in units of J_1 and δ the same gap in the scaled units in which the filters are designed; all fits and figures use Δ. To leave a fraction ϵ of excited weight in a trial state with ground-state overlap γ, a filter F must suppress the excited levels relative to the ground state by η with $\eta^2 \approx \gamma\epsilon/(1-\gamma)$.

SBC filter. One pulse [1] applies $\mathsf{H}_a\, e^{-itH_s \otimes Z_a}\, R_z^{(a)}(2\phi)\, \mathsf{H}_a$ to the system and an ancilla in $|0\rangle$; its ancilla-$|0\rangle$ block is $\cos(H_s t + \phi)$. Keeping outcome 0 after m pulses applies

$$F(H_s) = \prod_{i=1}^{m} \cos(H_s t_i + \phi_i), \qquad T = \sum_i |t_i|, \quad (1)$$

with success probability $p \approx \gamma F(0)^2$. We use two schedules. *Dyadic*: r repetitions of the halving ladder $t = \tau, 2\tau, \ldots, 2^{K-1}\tau$, $\tau = \pi/(1+\delta)$, $\phi = 0$. One block is a Dirichlet kernel,

$$\prod_{k=0}^{K-1} \cos(2^k E\tau) = \frac{\sin(2^K E\tau)}{2^K \sin(E\tau)}, \quad (2)$$

equal to 1 at $E = 0$ and at most $1/(2^K \sin\theta)$, $\theta = \pi\delta/(1+\delta)$, on $[\delta, 1]$, so $T \approx (e/\delta)\ln(1/\eta)$ in scaled time, the optimal scaling [6, 22], with no optimization and $F(0) = 1$. *Optimized*: SBC's energy recipe [1], times and phases that maximize $F(0)^2$ at fixed (T, m) subject to suppression η on the whole of $[\delta, 1]$; nonzero phases trade success probability for a shorter filter.

Trotterization. Each controlled evolution $e^{-it_i H_s \otimes Z_a}$ is approximated by $k_i = \lceil |t_i|/\delta t \rceil$ second-order (Strang) steps with a common step δt; the identity term becomes one exact ancilla R_z per pulse.

Gate counts. We assume all-to-all connectivity and Clifford+R_z gates. For a Heisenberg bond (i, j) the three commuting controlled terms form one exact block,

$$e^{-ict(X_iX_j + Y_iY_j + Z_iZ_j)\otimes Z_a} = V^\dagger e^{-ict(Z_iZ_a + Z_jZ_a - Z_iZ_jZ_a)} V,$$
$$V = \mathsf{H}_i\, \mathrm{CNOT}_{ij}, \quad (3)$$

whose diagonal factor is a phase polynomial on three parities [32]: 6 CNOTs and 3 R_z, against 12 CNOTs for three separate Pauli rotations. Control by an anticommuting Pauli string [8, 9] is not available, because no Pauli product anticommutes with all of XX, YY and ZZ on a bond. k Strang steps cost $k(2B-2)+1$ blocks for B bonds, and each pulse adds one ancilla R_z. An MPS of bond dimension 2^b is a staircase of $N - b$ unitaries on $b+1$ qubits, each costed by the quantum Shannon decomposition bound [13]. The cost of one attempt is the trial circuit plus the filter, and the expected cost per success is that divided by p (repeat-until-success).

Measured cost. For each case the exact filter is applied to the trial state to find the shortest filter that reaches infidelity ϵ (dyadic: every ladder (K, r); optimized: SBC's design with the true gap and overlap). At that filter, the coarsest common Strang step at which the full Trotterized circuit, simulated on the state vector, still reaches ϵ is found by bisection, and the cheaper of the two shortest dyadic ladders is kept. Optimized designs that barely pass the ground state ($p < \gamma/2$) are repeated over optimizer seeds (21 of 286 points). Every measured cost therefore uses the exact ground state: it is what the circuit has to do, not a rule an experiment could apply unchanged; Sec. VII plans from DMRG energies and a step certified on the device.

III. MODEL

We study the open spin-$\frac{1}{2}$ chain

$$H = J_1 \sum_{i=1}^{N-1} \mathbf{S}_i \cdot \mathbf{S}_{i+1} + J_2 \sum_{i=1}^{N-2} \mathbf{S}_i \cdot \mathbf{S}_{i+2}, \quad (4)$$

$J_1 = 1$, which is gapless at $J_2 = 0$ and dimerized above $J_2 \approx 0.2411$ [16, 17]; it has $B = 2N - 3$ bonds for $J_2 \neq 0$. We also study the same model on $4 \times L_x$ cylinders, periodic around the four-site circumference and open along L_x, at $J_2 = 0$ and 0.3, with the MPS following a snake path. Spectra are computed exactly per S^z block up to $N = 16$. MPS trials are the variationally optimal MPS at fixed bond dimension, computed with quimb [18]; for larger systems the energies and overlaps come from DMRG with bond dimension up to 256.

Why two dimensions. Figure 1 compares chains and cylinders at the same number of sites. Two differences favor a filter on cylinders. First, a snake MPS of fixed bond dimension is a far worse trial: at $N = 48$ an MPS with $\chi = 8$ leaves a residual weight $1 - \gamma = 10^{-2}$ on the chain but 0.69 on the 4×12 cylinder, and even $\chi = 64$ leaves 5.5×10^{-4} there. Reaching a tight target with the MPS alone therefore needs a large bond dimension, and its staircase circuit costs about four times more per doubling of χ (4 500, 19 536, 80 324 and 321 720 CNOTs for $\chi = 8$, 16, 32, 64 at $N = 48$). Second, the cylinder's gap stays open: it falls from 0.66 at 4×3 to 0.35 at 4×12, while the chain's falls to 0.08 at $N = 48$. The filter's depth is set by the gap and its overlap cost is only logarithmic (Sec. IV), so a large gap and a poor but cheap trial are the regime in which it should pay. Section VI tests this on the cylinders we can simulate.

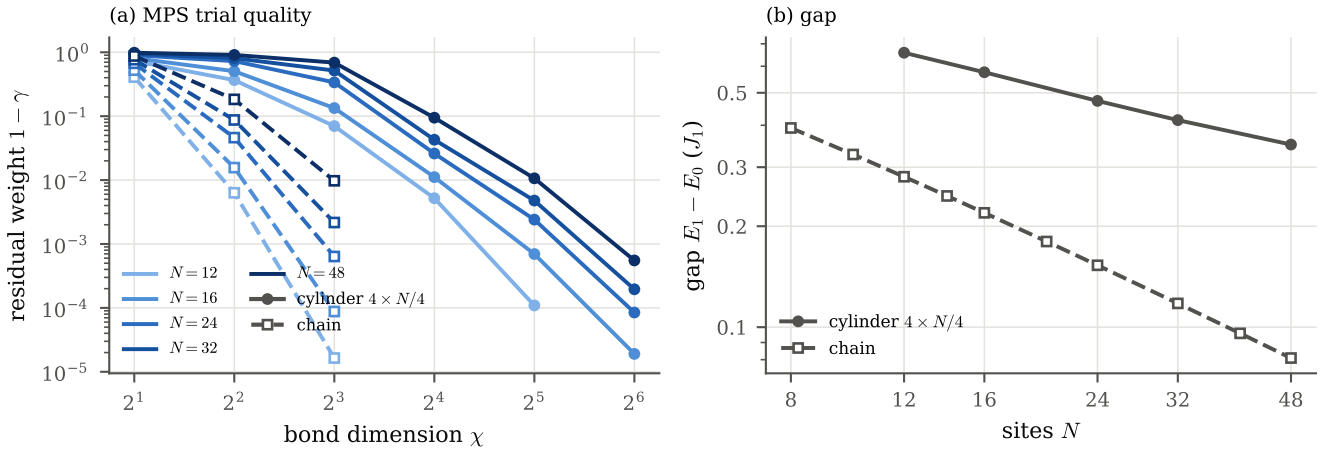


FIG. 1. Chains (open squares, dashed) and $4\times N/4$ cylinders (filled circles, solid) at $J_2=0$, from DMRG (verified against ED for $N\le 16$). (a) Residual weight $1-\gamma$ of the optimal MPS against its bond dimension; colors mark the number of sites. Chain trials were computed up to $\chi=8$. (b) Gap E_1-E_0 against N. Sources: `results/inputs.json`, `results/ext_inputs2d.json`.

IV. GAP AND OVERLAP AS INDEPENDENT INPUTS

Along any natural family of chains the gap and the overlap move together: at fixed J_2 the gap closes with N, and at fixed bond dimension the overlap decays with N. We therefore construct a family in which they are set independently at every N.

Construction. For each $N\in\{8,10,12,14\}$ and target gap $\Delta\in\{0.30,0.35,\ldots,0.55\}$, J_2 is found by bisection on exact diagonalization so that the singlet–triplet gap equals Δ to 10^{-9}; pairs without a solution below the dimerized regime are left out. For all tuned couplings the lowest $S^z=0$ excitation coincides with the lowest triplet to 10^{-7}, so Δ is the true gap. The trial is the variational $\chi=16$ MPS, prepared by its staircase circuit, followed by $R_y(\theta)$ on every even site; rotating one sublattice breaks the singlet symmetry and lowers γ smoothly, and θ is set by bisection to hit the target γ. The trial circuit is therefore the same along an overlap scan. Study A fixes $\Delta\in\{0.40,0.50\}$ and scans $\gamma\in\{0.03,0.1,0.3,0.5,0.7,0.9\}$; study B fixes $\gamma\in\{0.1,0.5\}$ and scans Δ; both at $\epsilon\in\{10^{-2},10^{-3},10^{-4}\}$, 222 cases in all, each with the dyadic and optimized schedules and with QETU (Sec. V). Costs are for the filter alone unless stated.

A. The gap sets the circuit, the overlap sets the repetitions

Figures 2 and 3 show CNOTs, R_z rotations and evolution time per shot, and CNOTs per success, along both scans.

Per shot, the overlap enters only logarithmically. At fixed gap, lowering γ thirtyfold (0.9 to 0.03) raises the CNOTs per shot of the dyadic filter by only 1.2–3.0× and its evolution time by 1.5–3.0×. The filter must reach $\eta^2=\gamma\epsilon/(1-\gamma)$, and a worse trial adds only $\ln 290\approx 5.7$ to $L=\ln[(1-\gamma)/(\gamma\epsilon)]$, already 2–13 here. The dyadic filter grows in whole pulses, so its per-shot cost moves in steps.

Per shot, the gap enters as a power. At fixed overlap, closing the gap by 1.8× (0.55 to 0.30) raises the CNOTs per shot by 2.7–4.3× and the evolution time by 1.5–2.8×. The evolution time follows the textbook $T=O(\Delta^{-1}\ln\eta^{-1})$ [5, 6] (fitted exponent 1.19); the CNOTs per shot grow faster, as $\Delta^{-1.91}$ (Table I), because a filter for a smaller gap also needs a finer Trotter step (Sec. IV D), and $1.8^{1.91}\approx 3$ matches the observed 2.7–4.3×.

Per success, the overlap returns as $1/\gamma$. Every run succeeds with probability $p\approx\gamma F(0)^2$, with $F(0)=1$ for the dyadic filter, so the cost per success carries $1/\gamma$. The gap decides the depth of the coherent circuit, and with it whether the filter fits a coherence or error budget at all; the overlap decides how often that circuit has to be run.

R_z counts follow CNOT counts. Each bond block has 3 R_z and 6 CNOTs and each pulse adds one rotation, so $R_z/\mathrm{CNOT}=0.50$ throughout, and every statement about CNOTs holds for the arbitrary rotations, and so for the T count of a fault-tolerant implementation up to the synthesis factor.

B. Cost laws with confidence intervals

We fit three costs to

$$C\approx a\,N^{\alpha}\,\Delta^{-\beta}\left[\ln(1/\epsilon)\right]^{\lambda}\gamma^{-\nu} \tag{5}$$

over all 222 cases: the filter's CNOTs per shot, its CNOTs per success, and the expected CNOTs per suc-

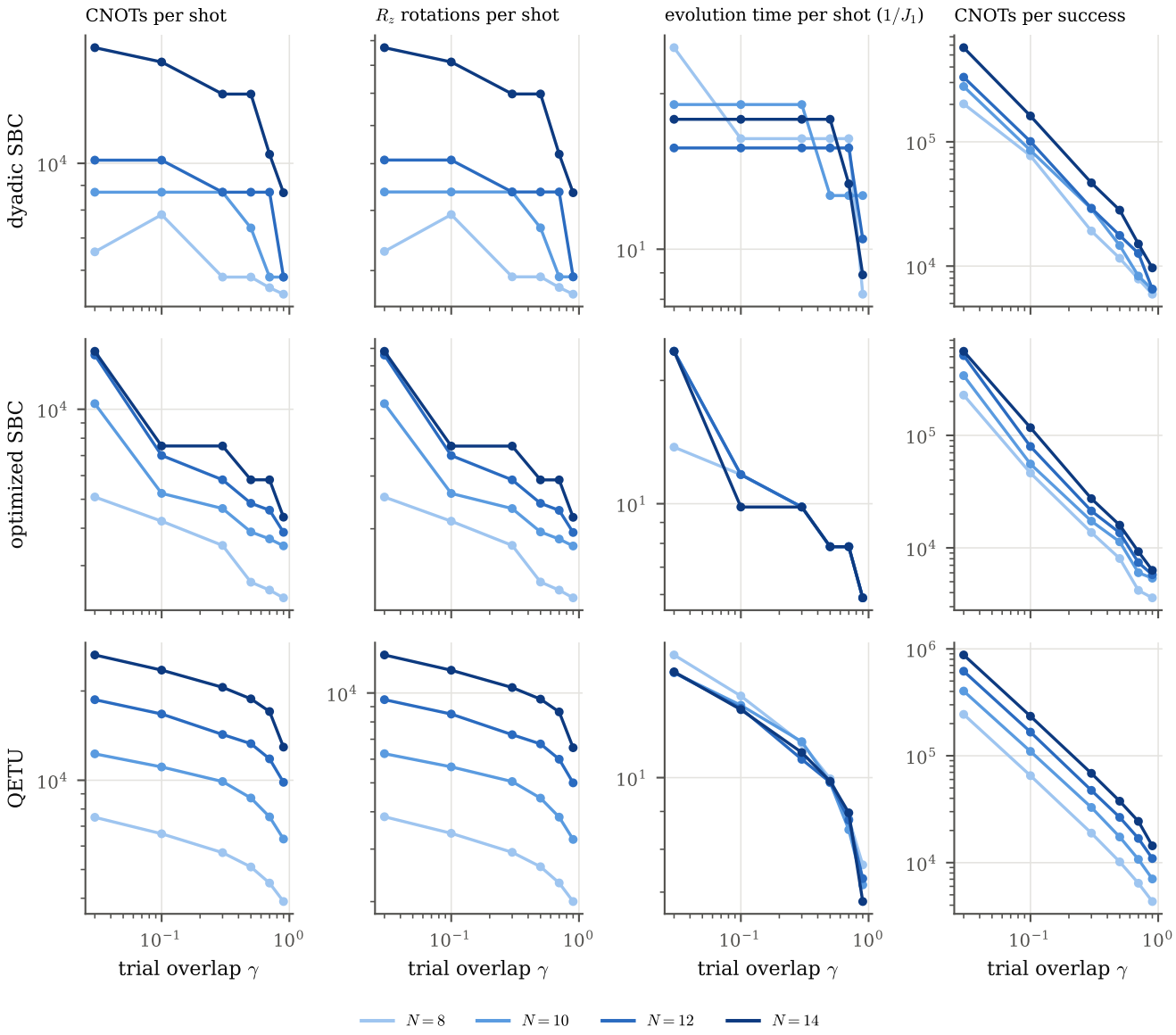


FIG. 2. Study A: filter-only resources at a gap held at $\Delta = 0.40$ for every N, $\epsilon = 10^{-3}$, against the trial overlap, measured by exact simulation. Rows: dyadic SBC, optimized SBC, QETU (Sec. V). Columns 1–3 per shot, column 4 per success. Source: `results/gapov.jsonl`, `results/gapov_filter.json`.

cess of the whole circuit (filter and trial). The target enters as $\ln(1/\epsilon)$, separately from the overlap, so that ν measures the overlap alone. The 95% intervals in Table I come from a bootstrap that resamples the 21 (N, Δ) systems, not the individual cases, since cases on one system share a Hamiltonian.

The table states the paper's main result with error bars. For the dyadic filter, per shot,

$$C_{\text{shot}} \propto N^{1.31}\, \Delta^{-1.91}\, \big[\ln(1/\epsilon)\big]^{2.40}\, \gamma^{-0.14}, \qquad (6)$$

with $\beta \in [1.60, 2.16]$ and $\nu \in [0.12, 0.16]$: the gap sets the circuit and the overlap barely enters it. Per success the overlap exponent becomes $\nu = 1.14$ $[1.11, 1.16]$, the $1/\gamma$ of repeat-until-success, and the gap exponent stays at 1.72. The optimized and Chebyshev filters show the same split. With the trial circuit included, a cost that does not depend on the gap, the gap exponent of the total falls to 1.19 [0.99, 1.37], which is why it is smaller than the per-shot value. Folding ϵ and γ into $L = \ln[(1-\gamma)/(\gamma\epsilon)]$, as the textbook bound does, gives an overlap exponent of 0.78 for the total cost, but only because L absorbs part of the $1/\gamma$; fitted separately, as here, the overlap exponent is consistent with 1. These exponents summarize $N = 8$–14 and $\Delta = 0.30$–0.55 and are not asymptotic laws. Appendix A compares the two SBC schedules and maps the cost over the (Δ, γ) plane and over (N, ϵ).

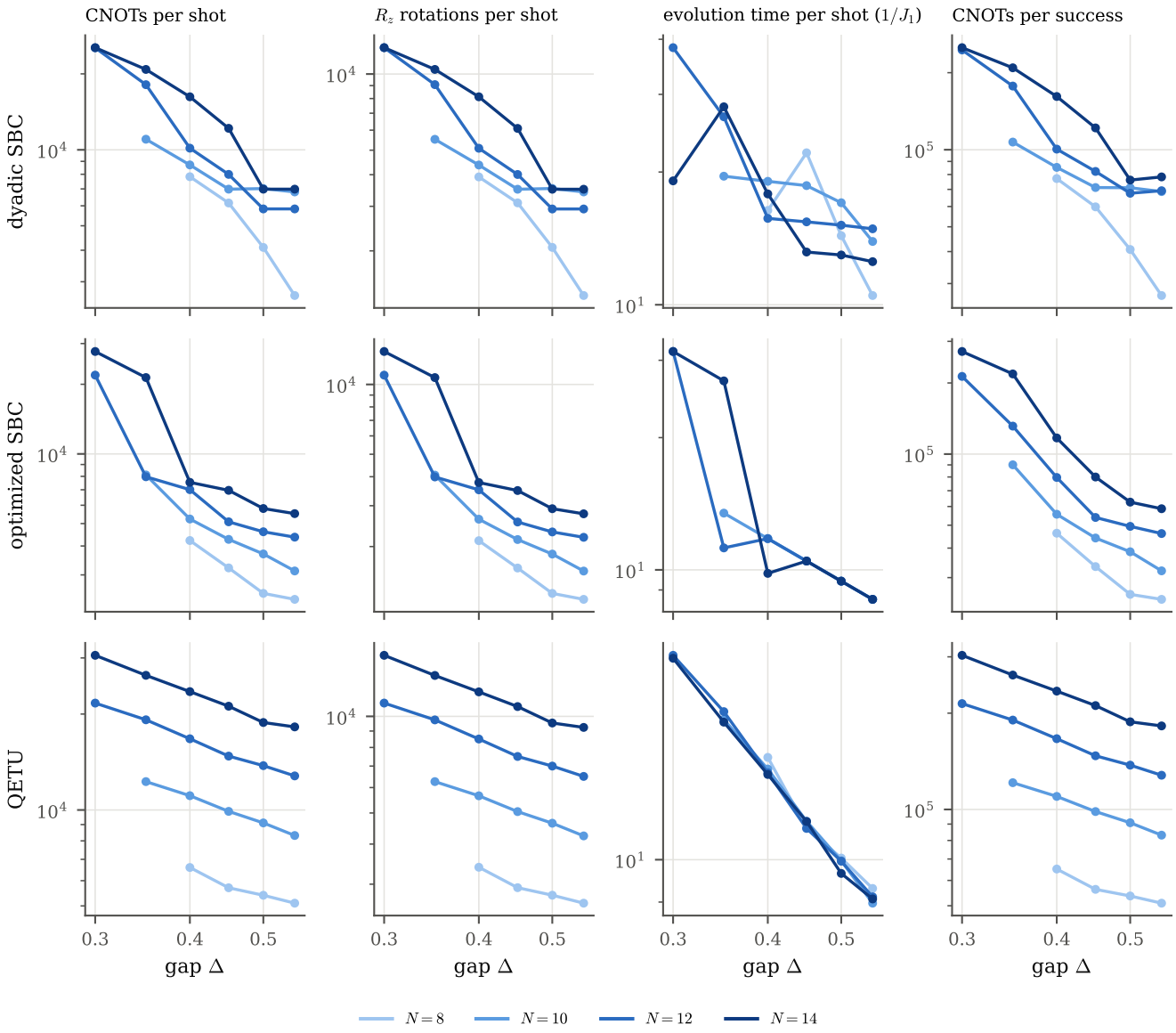


FIG. 3. Study B: as Fig. 2 at an overlap held at $\gamma = 0.10$ for every N, against the gap (held fixed across N by tuning J_2).

C. A second way to lower the overlap

The rotation lowers the overlap by spin flips, which put the missing weight mostly at high energy: the residual of a rotation trial has a mean excitation energy of 2–7 gaps. To check that the split between gap and overlap is not specific to that kind of residual, we repeat study A with a different mechanism. The trial is the exact ground state truncated to Schmidt rank one on a chosen set of bonds (at most five), an MPS of bond dimension at most 16 costed like the $\chi = 16$ trial, whose error is an entanglement truncation and whose residual sits low in the spectrum, 1.3–6.4 gaps above E_0. For each N, $\Delta \in \{0.40, 0.50\}$ and target γ the bond set whose overlap lies closest to the target is used, giving $\gamma = 0.03$–0.999 (79 cases with $1 - \gamma > \epsilon$; `run_review.py trunc`).

The same law holds. Fitting Eq. (5) to the truncation trials gives, for the dyadic filter, a per-shot overlap exponent $\nu = 0.28$ $[0.21, 0.36]$ and a per-success exponent 1.28 $[1.21, 1.37]$, against 0.14 and 1.14 for the rotation trials, with $\lambda = 2.38$ for the target. (The gap exponent is poorly constrained here, since only two gaps are used.) Predicting each truncation case's per-shot CNOTs from the rotation-trial fit of Table I is off by a median factor 0.90 (range 0.28–1.51). The overlap therefore enters through its size, not through where in the spectrum the missing weight sits: the filter has to suppress every level above the gap, wherever the weight is.

TABLE I. Fits of the measured costs to Eq. (5) over all 222 gap/overlap cases, with 95% intervals from a bootstrap over the 21 (N, Δ) systems. "rms": typical factor between a case and its fit. Source: `results/gapov.jsonl`; `make_review.py`.

filter	cost	a	α (N)	β (Δ^{-1})	λ $(\ln \epsilon^{-1})$	ν (γ^{-1})	rms
dyadic	filter, per shot	5.6×10^{-1}	$1.31_{[1.12,\,1.49]}$	$1.91_{[1.60,\,2.16]}$	$2.40_{[2.22,\,2.57]}$	$0.14_{[0.12,\,0.16]}$	1.24
dyadic	filter, per success	6.4×10^{-1}	$1.42_{[1.23,\,1.61]}$	$1.72_{[1.43,\,1.96]}$	$2.29_{[2.10,\,2.47]}$	$1.14_{[1.11,\,1.16]}$	1.25
dyadic	filter + trial, per success	5.3	$1.52_{[1.37,\,1.66]}$	$1.19_{[0.99,\,1.37]}$	$1.54_{[1.36,\,1.70]}$	$1.08_{[1.07,\,1.10]}$	1.19
optimized	filter, per shot	1.6×10^{-1}	$1.52_{[1.30,\,1.71]}$	$1.86_{[1.58,\,2.07]}$	$2.52_{[2.37,\,2.68]}$	$0.27_{[0.25,\,0.28]}$	1.26
optimized	filter, per success	2.5×10^{-1}	$1.45_{[1.27,\,1.63]}$	$2.01_{[1.76,\,2.19]}$	$2.44_{[2.29,\,2.60]}$	$1.20_{[1.18,\,1.21]}$	1.23
optimized	filter + trial, per success	4.6	$1.52_{[1.41,\,1.64]}$	$1.39_{[1.23,\,1.52]}$	$1.46_{[1.32,\,1.62]}$	$1.09_{[1.09,\,1.10]}$	1.20
QETU	filter, per shot	3.2	$2.24_{[2.11,\,2.32]}$	$0.99_{[0.87,\,1.08]}$	$0.88_{[0.85,\,0.90]}$	$0.18_{[0.18,\,0.19]}$	1.12
QETU	filter, per success	3.0	$2.24_{[2.12,\,2.33]}$	$0.98_{[0.86,\,1.07]}$	$0.89_{[0.87,\,0.92]}$	$1.18_{[1.17,\,1.19]}$	1.12
QETU	filter + trial, per success	1.1×10^{1}	$2.10_{[2.00,\,2.17]}$	$0.76_{[0.66,\,0.84]}$	$0.68_{[0.66,\,0.69]}$	$1.14_{[1.13,\,1.14]}$	1.08

D. Mechanism: filter length, Trotter step and gates per step

The per-shot cost factorizes as

$$\text{CNOTs per shot} = \frac{T}{\delta t} \times c_{\text{step}}(N), \tag{7}$$

and each factor answers to different inputs (Fig. 4).

Filter length. The physical evolution time depends on the gap and L only: a fit gives $T \propto \Delta^{-1.19} L^{0.77} N^{-0.24}$ for the dyadic filter, so at fixed absolute gap the filter length does not depend on the size of the system.

Trotter step. The filter tolerates very coarse steps: median $\delta t = 1.25$, 0.56, 0.30 (in $1/J_1$) at $\epsilon = 10^{-2}$, 10^{-3}, 10^{-4}, larger at larger gap. The step shrinks roughly as $\epsilon^{0.3}$ and as $\Delta^{0.7}$ (steps per shot grow as $\Delta^{-1.90}$ against $\Delta^{-1.19}$ for the time), and barely depends on N, so the number of steps per shot grows as $L^{0.71}$ and $N^{0.08}$. The bisection assumes that a circuit which passes at some step also passes at every finer one. We checked this on 36 cases ($N = 8$ and 10; three overlaps at $\Delta = 0.40$ and three gaps at $\gamma = 0.1$; all three targets) by scanning the common step on a dense logarithmic grid from one step per pulse down to an eighth of the bisected step. For the dyadic filter and for QETU (one to six steps per query) passing was monotone in every scan: once a step passes, every finer step passes. The exceptions are all in the optimized SBC filter, 66 of 72 SBC scans being monotone. Its pulse times are unequal, so the number of steps in each pulse is rounded separately and the realized error does not fall smoothly with the step. In four of the six exceptions the failing step lies above the bisected one, which only means the bisection could have stopped at a coarser step (the dense scan found a passing step up to 1.94 times coarser; median ratio 0.98). In two ($N = 8$ and 10 at $\Delta = 0.30$) a step finer than the bisected one fails, at 0.72 and 0.91 times it, with infidelity up to $2.05\,\epsilon$. The step we report is always one that was evaluated and passed, so the tables are unaffected, but for the optimized filter a finer step is not automatically safe and the chosen step must be checked directly.

Measured step against the bounds. Figure 5 sets each measured step beside two bounds evaluated for the same filter and target, with half of $\sqrt{\epsilon}$ allotted to Trotter error: the worst-case commutator bound for Strang steps (exact local norms of the nested commutators [30]) and the explicit low-energy bound of Mizuta and Kuwahara [31] (their Corollary S4), evaluated in its most favorable case, for a state inside the ground space. The measured dyadic step is a median 51 times the commutator step (range 8–152; 39 for the optimized filter, 14 for QETU, whose short queries leave less room). The low-energy bound, which improves on Şahinoğlu and Somma [19], scales better with the energy but does worse here: the ground state of a frustrated antiferromagnet lies an extensive energy above the zero of the positive-semidefinite bond terms that the bound measures from, and its constants restrict it to $\delta t \leq 0.0034/J_1$, so it asks for steps 242855 times finer than measured. Its asymptotic advantage does not reach these sizes. The pulses are long and smooth, the filtered state sits at the bottom of the spectrum, and the Trotter error that matters is the error on that state, which neither bound resolves.

Gates per step. One Strang step costs a median $20\,N$ CNOTs for the two-neighbor chain ($6(2B-2)$ with $B = 2N-3$ bonds). With a step count nearly independent of N, the CNOTs per shot grow as $N^{1.31}$, almost entirely through the gates per step.

V. COMPARISON WITH A CHEBYSHEV FILTER (QETU)

QETU [8] uses d queries of $e^{-i\tau H_s \otimes Z_a}$ interleaved with ancilla rotations and a single post-selection, so that the ancilla-0 block is $p(\cos \tau H_s)$ for a real even polynomial p of degree d with $|p| \leq 1$. With the stop band mapped to $[0, x_s]$, $x_s = \cos(\tau\delta)$, we take the extremal Chebyshev

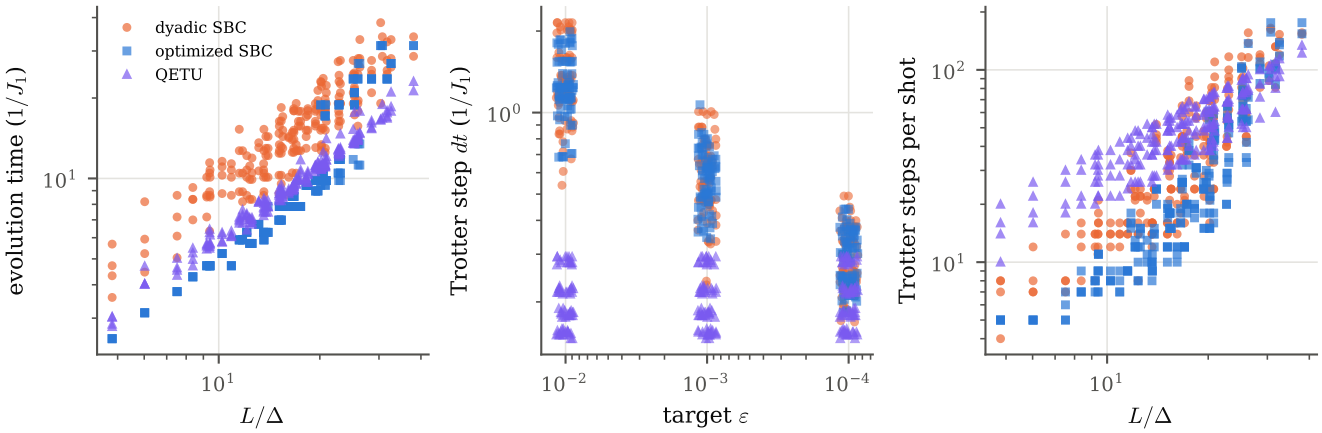


FIG. 4. All 222 cases, measured. (a) Evolution time against L/Δ, $L = \ln[(1-\gamma)/(\gamma\epsilon)]$. (b) Coarsest passing Trotter step against ϵ (jittered horizontally). (c) Trotter steps per shot against L/Δ. Orange circles: dyadic SBC; blue squares: optimized SBC; purple triangles: QETU.

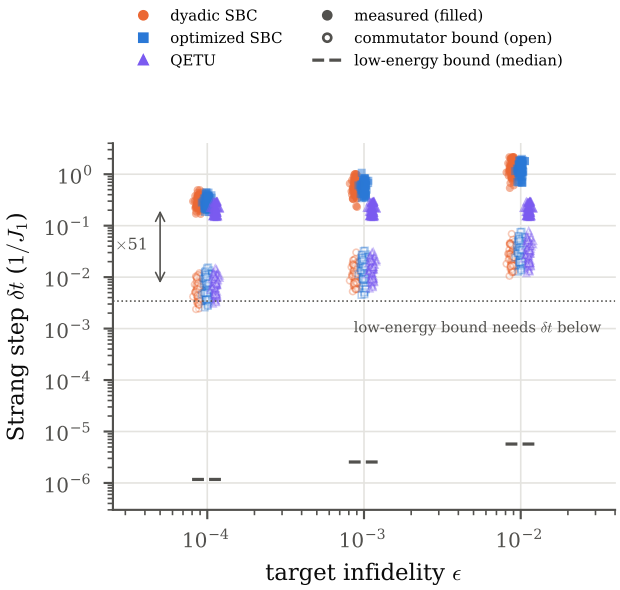


FIG. 5. Coarsest passing Strang step in the controlled study (filled) against the commutator bound for the same filter and target (open), and the median low-energy bound of Ref. [31] (dashed), which holds only below the dotted line. Source: `results/review_bounds.json`.

filter

$$p(x) = \frac{T_d(x/x_s)}{T_d(1/x_s)}, \tag{8}$$

which has p equal to 1 at the ground state and needs $T = d\tau \approx \delta^{-1}\ln(2/\eta)$, two to three times less evolution than the dyadic ladder. Each query is Trotterized like an SBC pulse and costed in the same controlled form (Sec. II); the classical phase-finding step costs no quantum gates and is not counted. By simulation on exact spectra for $N = 8$–16 (675 cases), the smallest degree that reaches ϵ is set by the gap and the target alone, $d_{\min}\delta \simeq (2/\pi)\cosh^{-1}(1/\eta)$, whatever the shape of the trial's excited weight, because the Chebyshev filter suppresses every level above the gap equally.

Query length $\pi/2$. With $\tau = \pi/2$, the choice of Dong, Lin and Tong [8], QETU is cheaper than the dyadic filter in 0 of 74 cases at $\epsilon = 10^{-2}$, 9 of 74 at 10^{-3} and 56 of 74 at 10^{-4}. The reason is the Trotter step, not the filter: QETU needs less evolution time (median 0.65× dyadic), but one Strang step per short query, $\delta t = \tau/W \propto 1/N$, so its step count is a median 1.40× the dyadic one and its CNOTs grow as $N^{2.22}$ against $N^{1.30}$. The long dyadic pulses tolerate steps of order $1/J_1$, but these shrink as roughly $\epsilon^{0.3}$, and by $\epsilon = 10^{-4}$ the shorter filter wins.

Longest admissible query. Any $\tau \le \pi/(1+\delta)$ keeps every excited level inside $|\cos\tau E| \le \cos\tau\delta$, and one Strang step per query still suffices at longer queries. Redoing every QETU search at $\tau = 3\pi/4$ and $\pi/(1+\delta)$ (467 cases: the 222 controlled cases and the 245 cases of Sec. VI) lowers its cost to a median 0.68 (range 0.29–4.28) of the $\tau = \pi/2$ cost at $\tau = \pi/(1+\delta)$. With QETU at its best τ (Table III) the SBC filters keep $\epsilon = 10^{-2}$ (QETU cheapest in 7 of 74), the two families split 10^{-3} (38 of 74), and QETU is the cheapest filter in 65 of 74 at 10^{-4}. The fair statement is that SBC wins at loose targets, the two are even at 10^{-3}, and the Chebyshev filter wins from 10^{-4} on.

The crossover moves with N. Because the QETU step shrinks as $1/N$ while the SBC step does not, the target at which QETU takes over should tighten with size. Table II shows that it does: the median cost of QETU at its best τ relative to the cheaper SBC filter, at fixed N, rises with N at $\epsilon = 10^{-2}$ (1.04 at $N = 8$ to 1.33 at $N = 14$) and stays near 0.83 at 10^{-4}, so the log-interpolated crossover moves from $\epsilon \approx 6.0 \times 10^{-3}$ at $N = 8$ to 7.3×10^{-4} at

TABLE II. Median cost of QETU (best τ) over the cheaper SBC filter in the controlled study, per N and target, and the target where the log-interpolated ratio crosses 1. Source: `results/qetu_tau.jsonl`, `make_review.py qcross`.

N	10^{-2}	10^{-3}	10^{-4}	crossover ϵ
8	1.04	0.89	0.85	6.0×10^{-3}
10	1.15	0.96	0.83	1.5×10^{-3}
12	1.30	1.01	0.83	7.3×10^{-4}
14	1.33	1.06	0.82	6.4×10^{-4}

TABLE III. Cheapest measured filter in the controlled study (74 cases per target; dyadic/optimized/QETU) with QETU at $\tau = \pi/2$ and at its best query length $\tau \in \{\pi/2, 3\pi/4, \pi/(1+\delta)\}$. Source: `results/qetu_tau.jsonl`.

target	$\tau = \pi/2$	best τ
$\epsilon = 1 \times 10^{-2}$	23/51/0	19/48/7
$\epsilon = 1 \times 10^{-3}$	16/57/1	4/32/38
$\epsilon = 1 \times 10^{-4}$	5/32/37	0/9/65

$N = 12$ and 6.4×10^{-4} at $N = 14$. Over this range the shift is slowing, consistent with the two cost exponents ($N^{2.22}$ against $N^{1.30}$) differing by roughly one power of N at loose targets and less at tight ones, but four sizes do not fix the asymptote. Chains of $N = 16$–24 at $\Delta = 0.40$ and two overlaps are being run with the same searches (`run_cluster.py chain`) to extend the table.

VI. BENCHMARK AGAINST THE MPS ALONE

The alternative to filtering a cheap trial is compiling a better MPS. We compare the two at equal target infidelity: for each system every MPS trial with $\chi = 2, \ldots, 64$ is decomposed on the exact spectrum, and for each $\epsilon \in [10^{-5}, 10^{-2}]$ the cheapest route is found, the smallest MPS that meets $1 - \gamma \le \epsilon$ alone, against the cheapest smaller MPS followed by a measured SBC or QETU filter, with cost (trial + filter)$/p$. Both routes use the staircase MPS cost of Sec. II.

Chains. On nine chains ($N = 12$, 14, 16 at $J_2 = 0$, 0.2411, 0.4; Fig. 6) compiling the MPS is cheaper in most cases: the filtered route wins 18 of 63 targets, by 1.1–3.2×. Every win filters a trial that misses the target by $(1 - \gamma)/\epsilon \le 8.8$, with a dyadic ladder of one to four pulses (1–21 Strang steps, $p \ge 0.988$), and all 90 candidate routes with a larger ratio lose, by a median 20×. The filter acts as a last-mile polisher: MPS cost jumps about fourfold per doubling of χ, so a target just below what a given χ reaches forces the MPS-alone route up a bond dimension, while a short filter removes the small residual of the lower one. The wins are therefore tied to the bond-dimension grid rather than to N or J_2. The optimized design never wins, because suppression over the whole band above δ needs $T \gtrsim \pi/\delta$ (scaled) where one dyadic pulse suffices, and QETU beats the MPS alone at 11 targets but is never the cheapest route.

Cylinders. On the 4×3 and 4×4 cylinders at $J_2 = 0$ and 0.3 (Fig. 7), the picture reverses, as Fig. 1 anticipates. A snake MPS struggles here: at 4×4, $J_2 = 0$, $\chi = 16$ misses $\epsilon = 10^{-2}$ ($1 - \gamma = 1.1 \times 10^{-2}$), $\chi = 32$ stalls at 7×10^{-4} and no $\chi \le 64$ reaches 10^{-5}, while the compile cost grows about fourfold per doubling (5 328, 20 548 and 76 600 CNOTs for $\chi = 16$, 32, 64). The filtered route is cheaper at 27 of the 28 cylinder targets, by 1.1–8.8×, and is the only MPS-based route at 10^{-5} on 4×4. Every win uses the dyadic ladder, mostly on a $\chi = 16$ trial, with $p \ge 0.77$ and 1–46 Strang steps, and recovers factors $(1 - \gamma)/\epsilon$ up to 10^3. The filter's depth grows only as $\ln(1/\epsilon)$ at fixed gap, while the MPS needed to reach ϵ alone grows fourfold per bond-dimension step. All trials are snake MPS, a weak choice in two dimensions, and the staircase cost charges every site a unitary on $b + 1$ qubits even where the bond is smaller, which favors the filtered route most where its trial is cheap.

Harder baselines. Two changes to the accounting test whether this conclusion rests on a weak baseline. First, an MPS staircase is a sequence of isometries rather than unitaries, and isometry decompositions save part of the quantum Shannon count [14]; charging every staircase unitary, in both routes, at half its quantum Shannon count leaves the filtered route cheaper at 13 of 63 chain targets (from 18) and 25 of 28 cylinder targets (from 27). Second, at these sizes the exact ground state is known, and generic state preparation [15] reaches any target at a fixed cost of 3 831, 15 511 and 62 423 CNOTs for 12, 14 and 16 qubits. It never beats the cheaper of the other two routes on chains. On the 12-site 4×3 cylinders it does, at 11 of their 14 targets ($\epsilon \le 3 \times 10^{-3}$), but already at 16 sites its cost exceeds every filtered route. Generic preparation needs the full state vector and grows as 2^N, so it is a ceiling only where exact diagonalization is possible; it shows that on the smallest cylinders the filtered route wins against the MPS, not against every alternative. A non-snake trial, a product of exact ground states of 4-site rings or 8-site blocks, and cylinders up to 4×6 are part of the larger runs described in Sec. VIII.

VII. PLANNING FROM DMRG INPUTS, VERIFIED AGAINST EXACT DIAGONALIZATION

The benchmarks above pick the filter and the Trotter step with the exact ground state. Here we run the workflow as an experiment would, with classical inputs from DMRG and a Trotter step chosen from quantities the device can measure, on systems small enough that exact diagonalization (ED) checks the outcome at the end. A

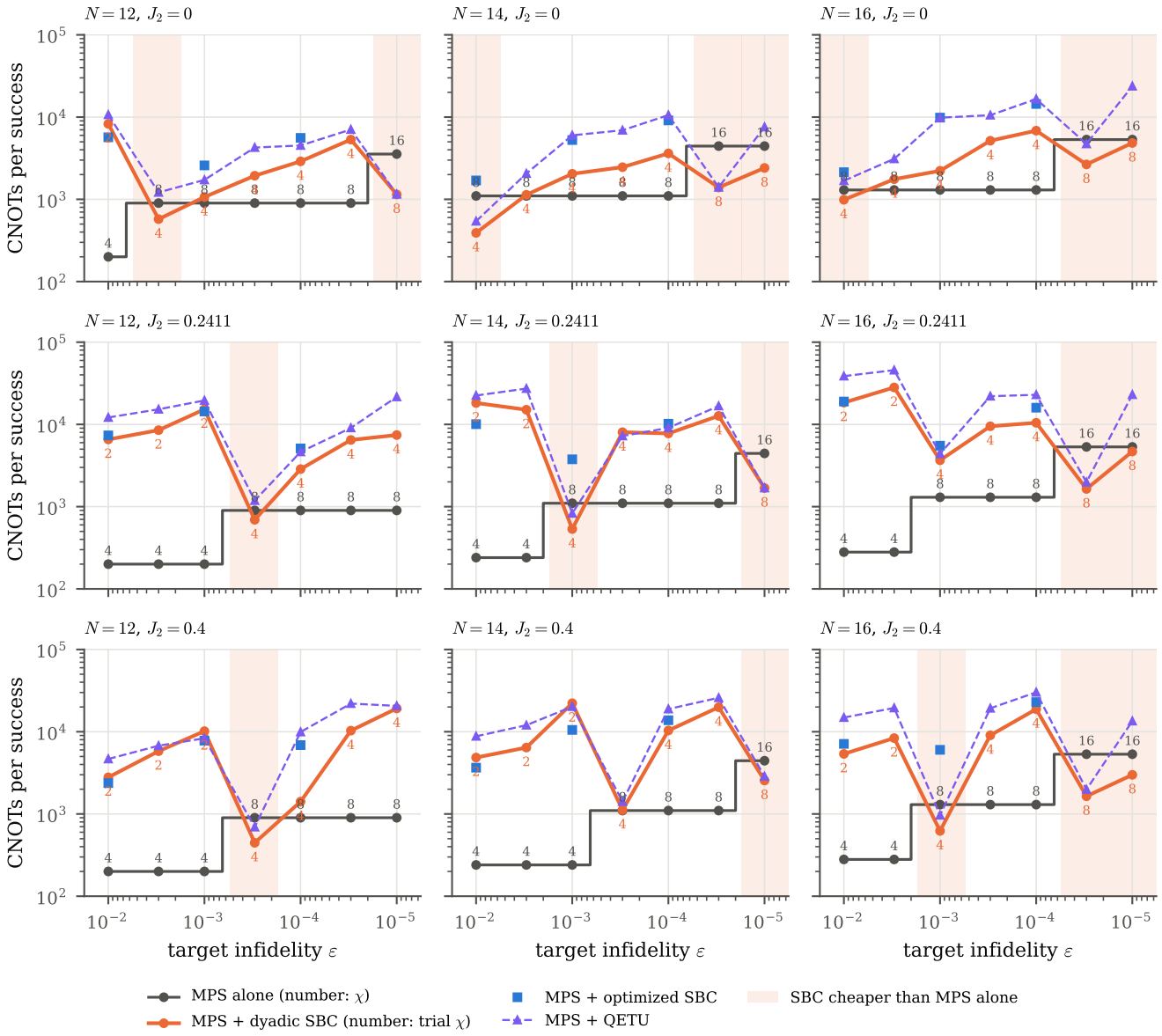


FIG. 6. Measured cheapest route to infidelity ϵ on chains of $N = 12$, 14, 16 (columns) at $J_2 = 0$, 0.2411, 0.4 (rows), all by exact simulation. Grey: cheapest MPS that meets $1 - \gamma \leq \epsilon$ alone (number: χ). Orange: cheapest smaller MPS followed by the dyadic SBC filter (number: trial χ); blue squares: optimized SBC (designed at $\epsilon = 10^{-2}, 10^{-3}, 10^{-4}$ only); purple: QETU. Shaded: the SBC route is cheaper. Cost is (trial + filter CNOTs)/p. Sources: `results/emp_actual1d.jsonl` and, for $N = 16$, $J_2 = 0.2411$, `results/emp_actual.json`.

large-bond-dimension DMRG run ($\chi \leq 256$) converges the energies to machine level: it reproduces the ED values of E_0 and E_1 (the lowest $S^z = 1$ state) to 2.3×10^{-11} and 2.4×10^{-11} at every size checked, and it provides a reference MPS. The trial that is compiled into the circuit is a much smaller MPS, and its overlap γ with the reference MPS, which agrees with its overlap with the ED ground state to 4.0×10^{-8}, is what falls short of the target. The energies place the filter's window; the overlap is the reason a filter is needed and sets how much it must suppress.

A step the device can certify. The Trotter step cannot be taken from DMRG. With exact E_0 and E_1, the Eckart bound

$$1 - F \leq \frac{\langle H \rangle - E_0}{E_1 - E_0} \tag{9}$$

turns the post-selected energy $\langle H \rangle$, which the device measures, into a certificate on the infidelity $1 - F$. The rule is therefore: starting from $\delta t = 4/J_1$, halve the step until the measured right-hand side of Eq. (9), with the DMRG energies, is at most ϵ. The bound charges every

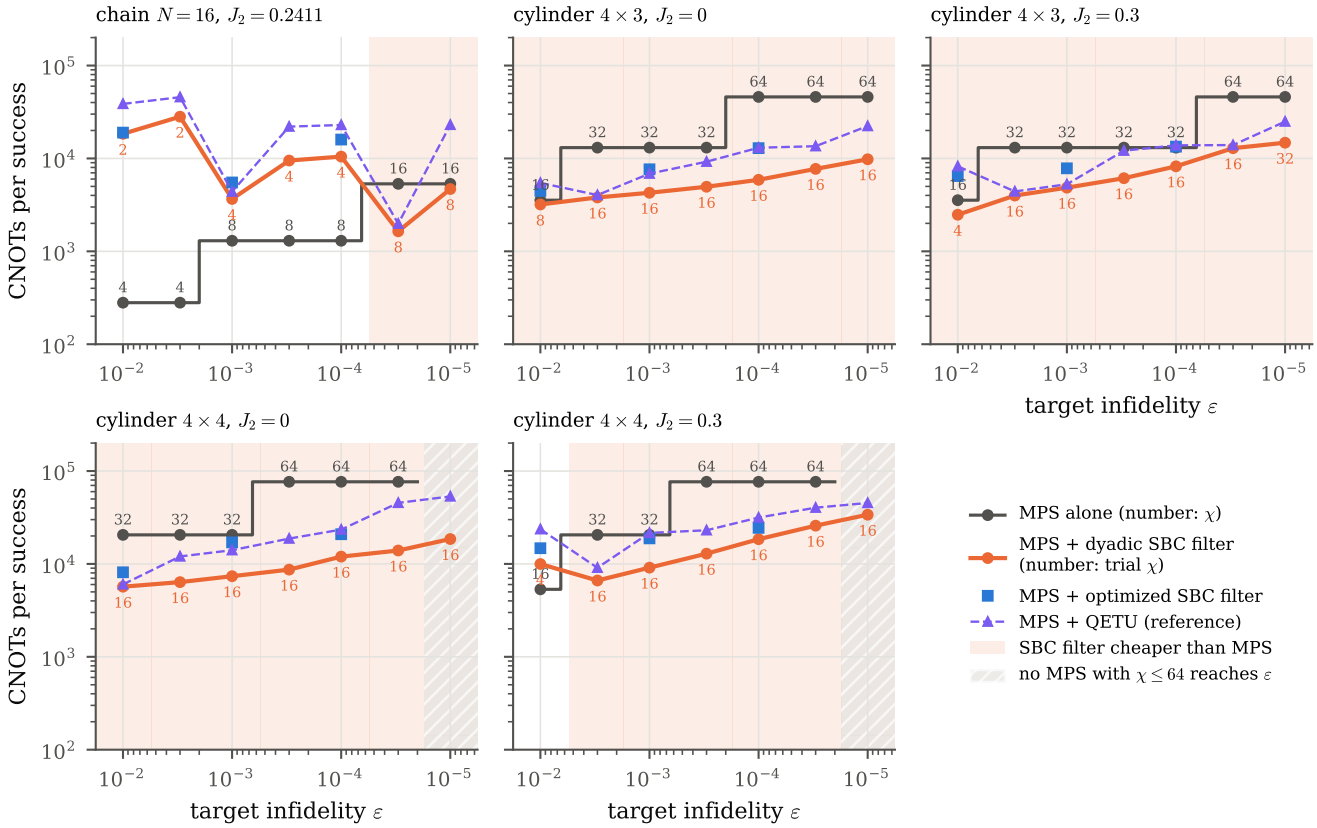


FIG. 7. Measured cheapest route to infidelity ϵ on the 16-site chain and on small cylinders. Grey: the cheapest MPS that meets the target alone (number: χ). Orange: the cheapest smaller MPS followed by the dyadic SBC filter (number: trial χ); blue squares: optimized SBC; purple: QETU. Shaded: the filtered route is cheaper; hatched: no MPS with $\chi \leq 64$ reaches ϵ. Every cost is (trial + filter CNOTs)/p from exact simulation. Source: `results/emp_actual.json`.

leaked level as if it sat at E_1 only in the best case; weight higher in the spectrum counts more, so a filter designed to leave exactly ϵ may never certify. In that case the filter is redesigned for $\epsilon/2$, $\epsilon/4$, ..., until one certifies (`run_review.py eckart`).

End to end. For the chain ($N = 12$, 14) and 4×3 cylinder targets of Sec. VI, $\epsilon = 10^{-5}$–10^{-2} (the $N = 16$ and 4×4 targets are still running), with the SBC (dyadic or optimized) and QETU filters (91 cases), the trial is the low-χ MPS ($\chi = 2$–64) that leads to the cheapest circuit. Its residual weight $1 - \gamma$ ranges from 2×10^{-5} to 0.4 and exceeds the target by a median factor of 7. The filter is designed from the DMRG energies and the DMRG estimate of γ. The rule certified 91 of the 91 plans, 25 of them after a redesign, and ED confirms that every certified state met the target (91 of 91), as Eq. (9) guarantees. The certificate costs a median 1.3 (SBC) and 1.3 (QETU) times the gate cost with the coarsest step that passes, since it accepts only steps fine enough to satisfy a bound looser than the true infidelity. An earlier rule, halving the step until the energy stops changing by $\epsilon\Delta/2$, met the target in all 182 cases at 1.9 and 1.9 times that cost, but carries no guarantee. The price of either rule is calibration: certifying Eq. (9) to two standard errors in the three measurement bases of the Heisenberg energy takes a median 2.7×10^{10} post-selected shots per case, growing as ϵ^{-2}, which pays off only if the circuit is then reused many times. The workflow therefore needs DMRG and an on-device calibration, not DMRG alone.

Sensitivity to the inputs. With converged DMRG the energies are not the limiting input, but a filter should still fail gracefully if they are off, for example when DMRG is not converged on a harder system. We therefore perturbed each input for every $N \leq 14$, all three couplings and the dyadic, optimized and QETU filters at $\epsilon = 10^{-2}$, with a conservative Trotter step so that only the filter design is tested, and simulated each circuit on the true trial state (Table IV). With exact inputs all 36 filters meet the target. Two inputs can break a filter, the gap and the overlap, and both only when overestimated.

Gap. Overestimating the gap moves the stop-band edge above the first excitation: by 10% it breaks the target in 2 of 36 cases (marginally, worst 1.1×10^{-2}), by 25% in 10 and by 50% in 16. The dyadic filter is the most tolerant (0 misses at 25%), because its suppression is far below the target over most of the band; the optimized

and Chebyshev filters, designed to sit right at the target, have no slack. Underestimating the gap is safe: assuming 0.5 of it costs 2.9×. The other two energies did not break any plan. Shifting the assumed ground energy by $\pm 0.1\Delta$ or $\pm 0.25\Delta$, with E_1 held at its true value, left 0, 0, 0 and 0 misses of 36 plans (12 per filter). The stop band starts at the assumed E_1, so an error in E_0 alone only moves the pass band, and a ground state a quarter of a gap from its assumed position is still passed with negligible loss; what breaks a filter is an overestimated E_1, which is what the gap rows above test. Placing $E_{\max}$ 5% or 10% below the true top, so that the highest levels fall outside the window, gave 0 and 0 misses of 36, because those levels carry almost no weight of an MPS trial; the default Weyl bound lies above the top in any case.

Overlap. Overestimating γ by 2–5% broke the target in 12 of 72 cases, every one because the inflated overlap crossed $1-\epsilon$ and the pipeline dropped the filter altogether. A safety margin removes this failure: designing for the assumed γ divided by 1.1 left 0, 0 and 0 misses of 36 at 2%, 5% and 10% overestimates. When DMRG is not converged, its error bars catch it: taking each energy's change between the last two bond dimensions as its error, the intervals contained both exact energies in all 75 tests on stopped bond-dimension ladders (`run_margin.py`), and the energy window became inconsistent, so that the pipeline refused to plan, before the gap error grew large enough to matter.

VIII. DISCUSSION

What sets the cost. The gap determines the coherent circuit: evolution time $\propto L/\Delta$, independent of N at fixed gap, and CNOTs and R_z per shot that scale as a power of $1/\Delta$. The overlap enters the circuit only through a logarithm and the expected cost through $1/\gamma$, so the filter tolerates a poor trial by paying in repetitions rather than depth; amplitude amplification [33, 34] would turn $1/\gamma$ into $1/\sqrt{\gamma}$ at the price of a deeper coherent circuit, and is not modeled. The target enters through L and, because the tolerated Trotter step shrinks as $\epsilon^{0.3}$, through the step count.

Which filter. The long, smooth SBC pulses tolerate Trotter steps of order $1/J_1$, which is why the cosine filter wins at loose targets even though it needs more evolution time than a Chebyshev filter; from $\epsilon = 10^{-4}$ the shorter Chebyshev filter wins. Among the SBC schedules the optimized design is cheaper in most cases (median 0.89× dyadic) but needs a nonconvex design that occasionally fails, while the dyadic ladder needs no design at all.

When filtering pays. A filter pays where the trial is hard to improve. For chains the MPS infidelity falls quickly with χ, so the filter helps only as a last-mile polisher; on small cylinders, where a snake MPS needs a fast-growing bond dimension, it wins almost everywhere. DMRG on larger cylinders, up to 4×12, shows the same two conditions growing stronger with size: the snake MPS gets worse at fixed χ while the gap stays above 0.35. That suggests two-dimensional systems as the place to look for an advantage, but it is a conjecture, not a result: the baseline is a snake MPS costed by a generic bound, a better trial (another ordering, an isometric or 2D tensor network) would narrow the gap, and the filtered cost beyond 16 sites is not measured here. Measured searches on 4×5 and 4×6 cylinders with ring and block orderings (`run_cluster.py cyl`) are the next test.

Measured costs and what an experiment pays. The benchmarks use the exact ground state to pick the shortest sufficient filter and the coarsest sufficient Trotter step. Planned from DMRG energies, with the step certified by the measured energy through Eq. (9), the filter met the target in every case we could check, at about 1.3 times the measured gate cost, but the certificate costs $\propto \epsilon^{-2}$ calibration shots, so the workflow needs DMRG and the device, not DMRG alone. Converged DMRG makes the energies essentially exact, and the small trial MPS's overlap is what the filter has to repair. If the energies were off, the gap is the input to protect: a 10–25% overestimate breaks the optimized and Chebyshev filters, while an underestimate costs little, and an overestimated overlap is absorbed by designing for $\gamma/1.1$. All of this was verified at sizes where ED is available; at larger sizes the same workflow runs, but its outcome can no longer be checked.

Limitations. Noiseless simulation; all-to-all connectivity; repeat-until-success rather than early abort; trial circuits costed by a generic decomposition bound; and sizes ($N \leq 16$, 16-site cylinders) at which classical methods are exact, so the fitted exponents are descriptive. The gap range that can be held fixed across N (0.30–0.55) lies in the easy regime, well above the gaps of longer natural chains.

CODE AND DATA AVAILABILITY

The `sbpipe` package, the study scripts, all raw results and the sources of this paper are available at https://github.com/aaronseymour7/FilterPipeline.

Appendix A: Dyadic versus optimized SBC, and cost maps

The dyadic ladder has zero phases, so it passes the ground state with $F(0) = 1$ and its length grows in whole pulses. The optimized design is shorter, at the price of $F(0) < 1$: its success probability is a median 0.88 times γ (as low as 0.52). Per shot it uses a median 0.58 times the dyadic evolution time and 0.62 times its Trotter steps, at the same tolerated step. Per success it is a median 0.89× the dyadic cost (range 0.58–1.55) and the cheaper one in 169 of 222 cases. Counting the cheapest of the three filters case by case, the optimized SBC filter wins

TABLE IV. Wrong inputs for every $N \in \{8, 10, 12, 14\}$ and J_2 (12 instances), $\epsilon = 10^{-2}$, second order: number of plans that miss the target and the worst measured infidelity. f: assumed over true value, except for E_0, where it is the shift in units of the gap; superscripts count plans the pipeline refused. Sources: `results/ext_sensitivity.json`, `results/review_sens2.json`.

		dyadic		optimized		QETU	
input	f	miss	worst	miss	worst	miss	worst
gap	1	0/12	1.7×10^{-3}	0/12	4.2×10^{-3}	0/12	2.3×10^{-3}
gap	1.1	0/12	6.5×10^{-3}	1/12	1.1×10^{-2}	1/12	1.0×10^{-2}
gap	1.25	0/12	6.8×10^{-3}	5/12	2.6×10^{-2}	5/12	3.0×10^{-2}
gap	1.5	2/12	6.3×10^{-2}	7/12	6.8×10^{-2}	7/12	9.2×10^{-2}
gap	2	8/12	1.4×10^{-1}	7/12	1.5×10^{-1}	11/12	2.2×10^{-1}
gap	3	11/12	2.2×10^{-1}	12/12	2.5×10^{-1}	12/12	3.1×10^{-1}
E_0 shift / Δ	-0.25	0/12	1.3×10^{-3}	0/12	5.5×10^{-3}	0/12	3.7×10^{-3}
E_0 shift / Δ	-0.1	0/12	2.4×10^{-3}	0/12	4.5×10^{-3}	0/12	2.8×10^{-3}
E_0 shift / Δ	+0.1	0/12	1.1×10^{-3}	0/12	3.6×10^{-3}	0/12	4.3×10^{-3}
E_0 shift / Δ	+0.25	0/12	3.5×10^{-3}	0/12	3.0×10^{-3}	0/12	4.8×10^{-3}
$E_{\max}$ / exact	0.9	0/12	1.9×10^{-3}	0/12	4.2×10^{-3}	0/12	2.5×10^{-3}
$E_{\max}$ / exact	0.95	0/12	1.1×10^{-3}	0/12	4.2×10^{-3}	0/12	3.6×10^{-3}
$E_{\max}$ / exact	1	0/12	1.7×10^{-3}	0/12	4.2×10^{-3}	0/12	2.3×10^{-3}
$E_{\max}$ / exact	1.25	0/12	1.1×10^{-3}	0/12	4.2×10^{-3}	0/12	2.9×10^{-3}
γ, no margin	1.02	1/12	2.4×10^{-2}	1/12	2.4×10^{-2}	1/12	2.4×10^{-2}
γ, no margin	1.05	3/12	5.5×10^{-2}	3/12	5.5×10^{-2}	3/12	5.5×10^{-2}
γ, margin 1.1	1	0/12	2.1×10^{-3}	0/12	3.1×10^{-3}	0/12	1.8×10^{-3}
γ, margin 1.1	1.02	0/12	2.1×10^{-3}	0/12	3.3×10^{-3}	0/12	3.7×10^{-3}
γ, margin 1.1	1.05	0/12	2.1×10^{-3}	0/12	3.4×10^{-3}	0/12	3.1×10^{-3}
γ, margin 1.1	1.1	0/12	1.7×10^{-3}	0/12	4.2×10^{-3}	0/12	2.3×10^{-3}

51 of 74 cases at $\epsilon = 10^{-2}$ (dyadic the other 23), 57 at 10^{-3} (dyadic 16, QETU 1), and 32 at 10^{-4}, where QETU takes 37 and dyadic 5, all with QETU at $\tau = \pi/2$. The price of the optimized schedule is its design: a nonconvex optimization of about 40 s per case that had to be repeated with other seeds in 21 of 286 points.

At fixed N and $\epsilon = 10^{-2}$ (Fig. 8), the contours of the dyadic cost per success run nearly parallel to the gap axis for $g \gtrsim 0.40$: there the overlap alone sets the cost, from 2–6 $\times 10^3$ CNOTs at $\gamma = 0.9$ to 1–4 $\times 10^5$ at $\gamma = 0.03$; below $g \approx 0.40$ the gap starts to matter. The optimized filter is cheaper over most of the map, and its contours tilt toward the gap axis, as its steeper gap exponent implies. At fixed gap and overlap (Fig. 9), the cost per success grows 7–21× (dyadic) from $\epsilon = 10^{-2}$ to 10^{-5} and 1.8–4.6× from $N = 8$ to 14, so over this range the target matters more than the size.

Appendix B: Reproducing the results

The code, raw results and paper sources are at `https://github.com/aaronseymour7/FilterPipeline`. The controlled study is run by `sbpipe_study/run_gapov.py` and summarized by `derive_gapov_filter.py`; the benchmarks against the MPS alone and the QETU degree analysis by `run_empirical.py`; the QETU query-length study by `run_qetu_tau.py`; the DMRG-only runs by `run_aposteriori.py`, the DMRG error-bar test by `run_margin.py`, and the wrong-input study by `run_extended.py`. `make_figures.py` writes the tables and numerical macros, and `make_figures_measured.py` the measured-only figures and fit table used here. `python -m pytest sbpipe/tests` checks the components, including the bond block of Eq. (3) and the Dirichlet identity (2).

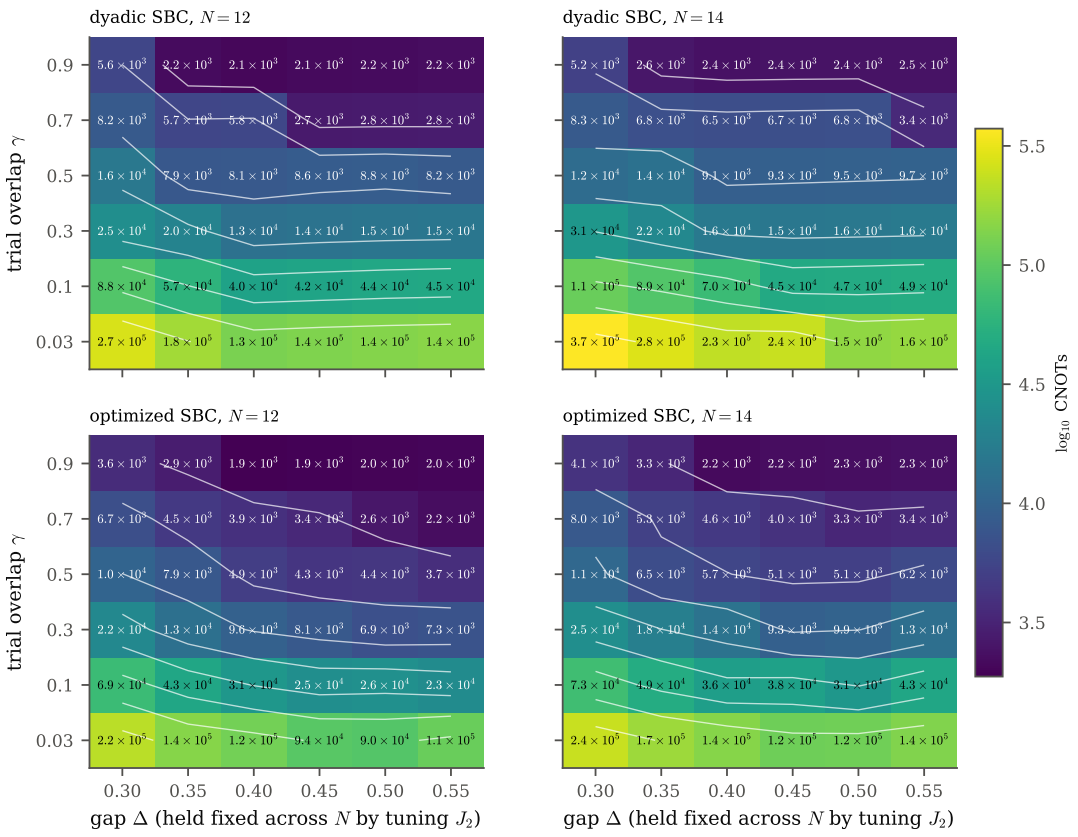


FIG. 8. Filter CNOTs per success over gap and overlap at $\epsilon = 10^{-2}$, measured, for the dyadic (top) and optimized (bottom) SBC filters at $N = 12$ and 14. White contours every factor 1.8. Source: `results/gapov.jsonl`, `results/gapov_grid.jsonl`.

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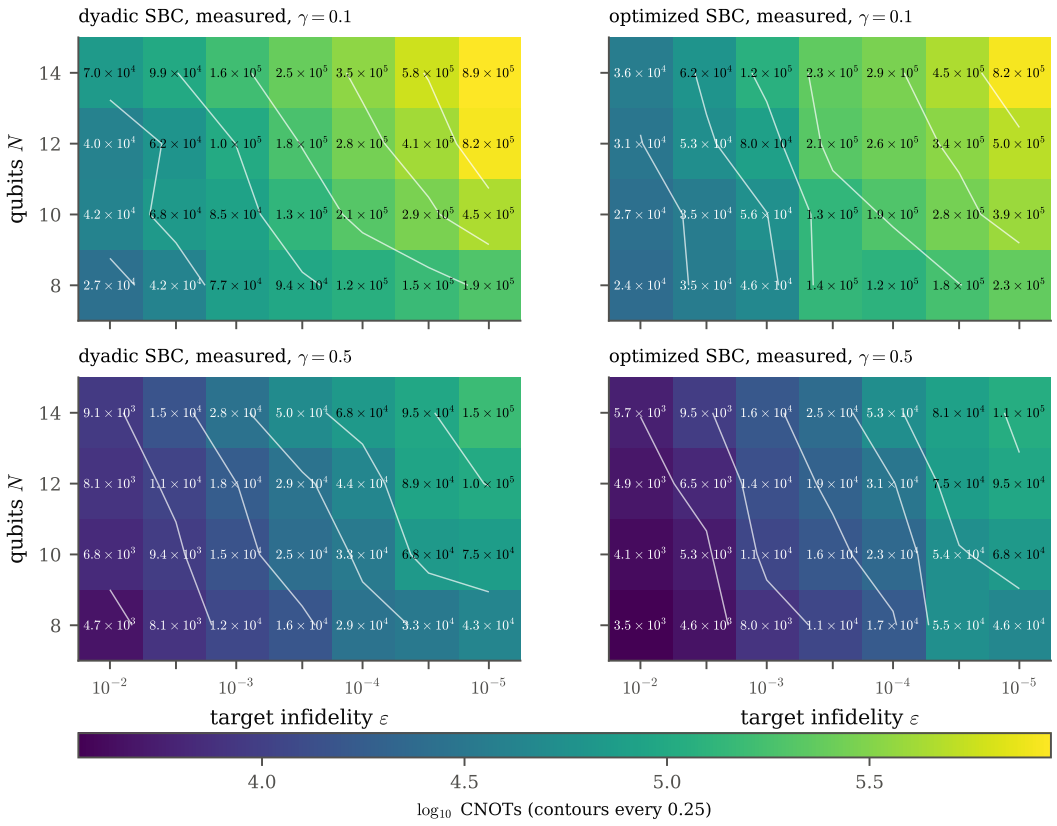


FIG. 9. Filter CNOTs per success over N and ϵ at a gap held at $g = 0.40$, measured, for the dyadic (left) and optimized (right) SBC filters at $\gamma = 0.1$ (top) and 0.5 (bottom). Source: `results/gapov*.jsonl`.

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